Cell Society Project Design

Michael, Joseph, Georgiy

Simulation API

Key Abstractions

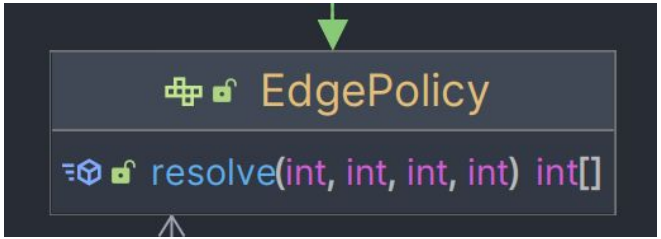




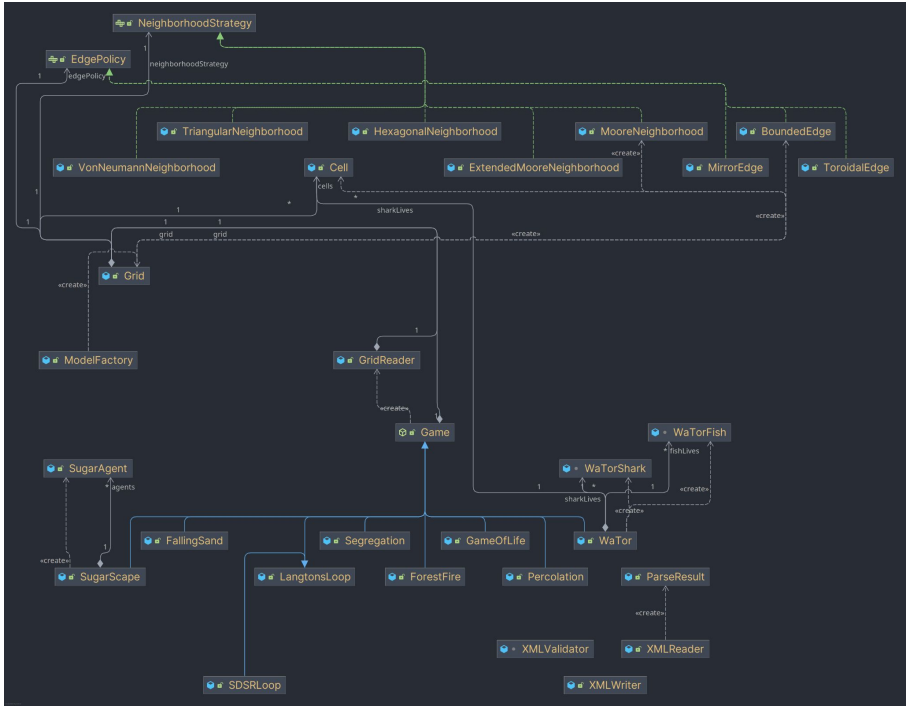
Simulation API

Further Abstractions





Simulation API



Simulation API

Use Case:

```
@Override
protected void updateGame() {
  int rows = getGrid().getRows();
  int cols = getGrid().getColumns();
  int iterations = rows * cols;
  for (int i = 0; i < iterations; i++) {
    int r = random.nextInt(rows);
    int c = random.nextInt(cols);
    int state = getGrid().getCell(r, c).getState();
    if (state == SAND) {
      moveSand(r, c);
    } else if (state == WATER) {
      moveWater(r, c);
    }
  }
}
```

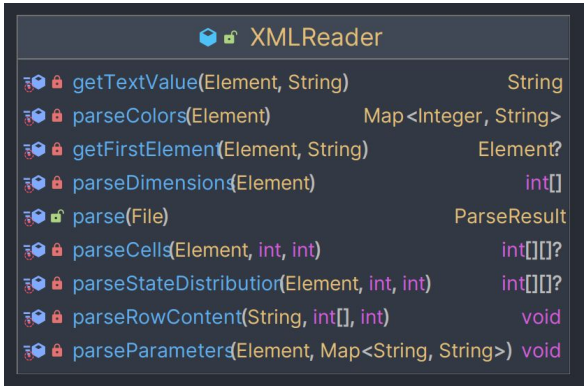
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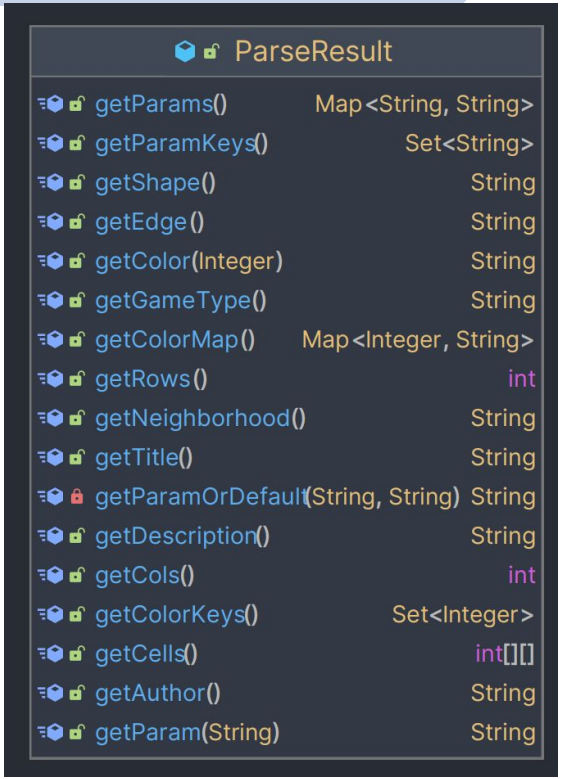
```
@Override
protected void updateGame() {
    for (int i = 0; i < getGrid().getRows(); i++) {
        for (int j = 0; j < getGrid().getColumns(); j++) {
            Cell cell = getGrid().getCell(i, j);
            int aliveNeighbors = countNeighborsWithState(get

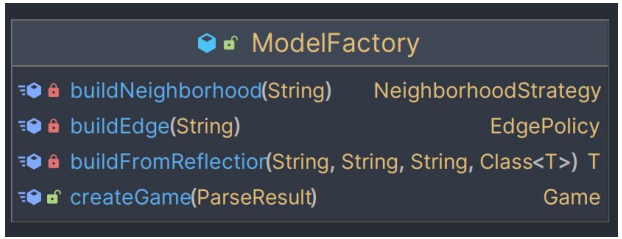
            if (cell.getState() == ALIVE) {
                cell.setNextState(survivalCounts.contains(al
            } else {
                cell.setNextState(birthCounts.contains(alive
            }
        }
    }
}
```

Configuration API

Key Abstractions







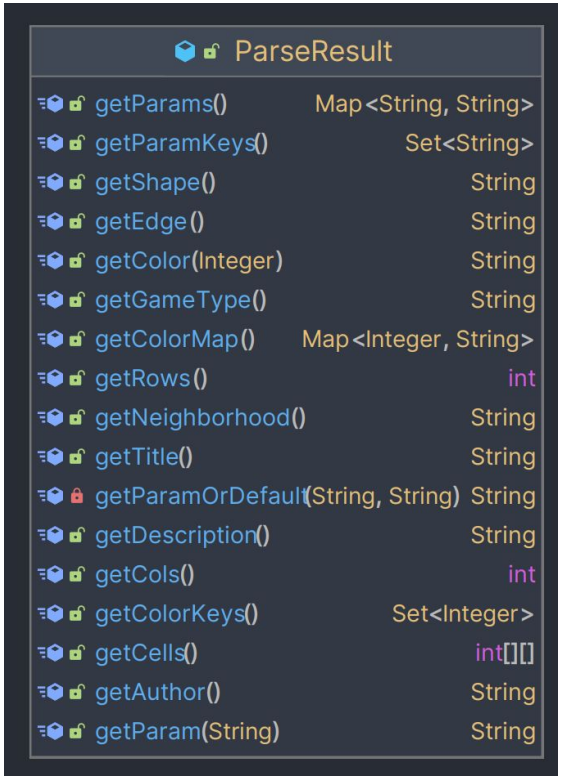


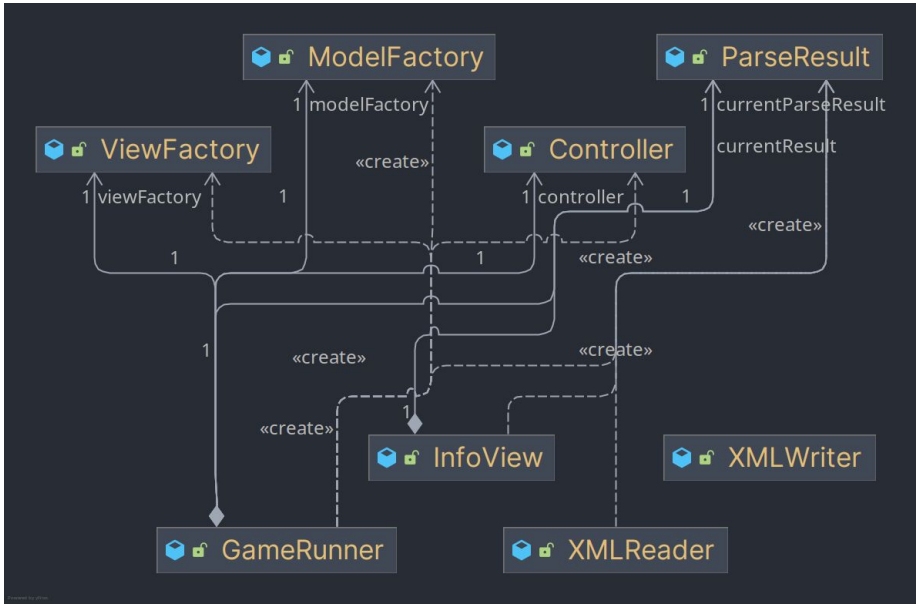
Configuration API

Use Case

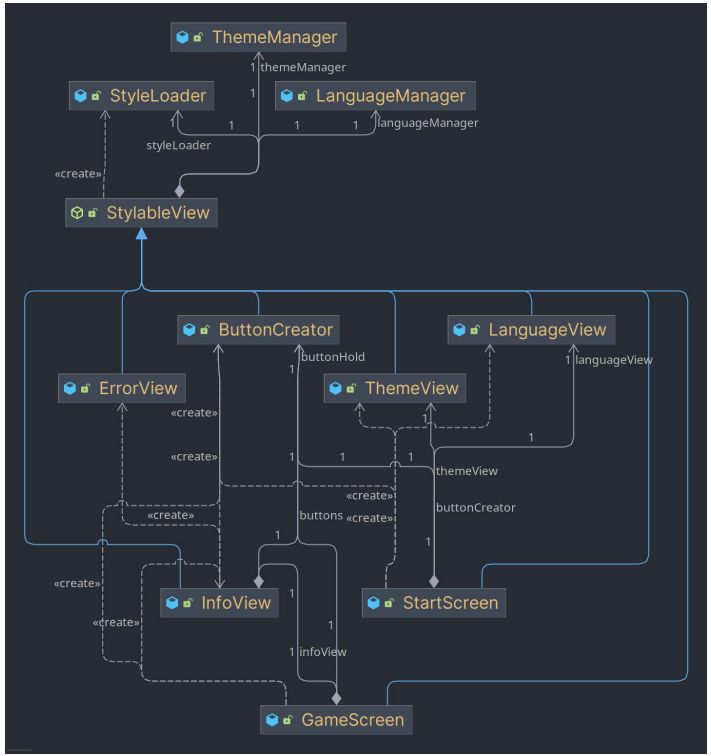
```
private <T> T buildFromReflection(String basePackage, String name,
    Class<T> expectedType) {
  try {
    Object instance = Class.forName(basePackage + name + suffix)
        .getDeclaredConstructor().newInstance();
    return expectedType.cast(instance);
  } catch (Exception e) {
    String message = name + " constructor could not be invoked";
    LOG.warn(message, e);
    throw new IllegalArgumentException(message, e);
  }
}
```

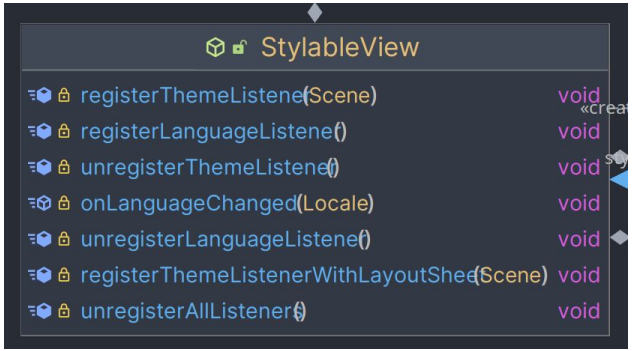
Stable Design - ParseResult





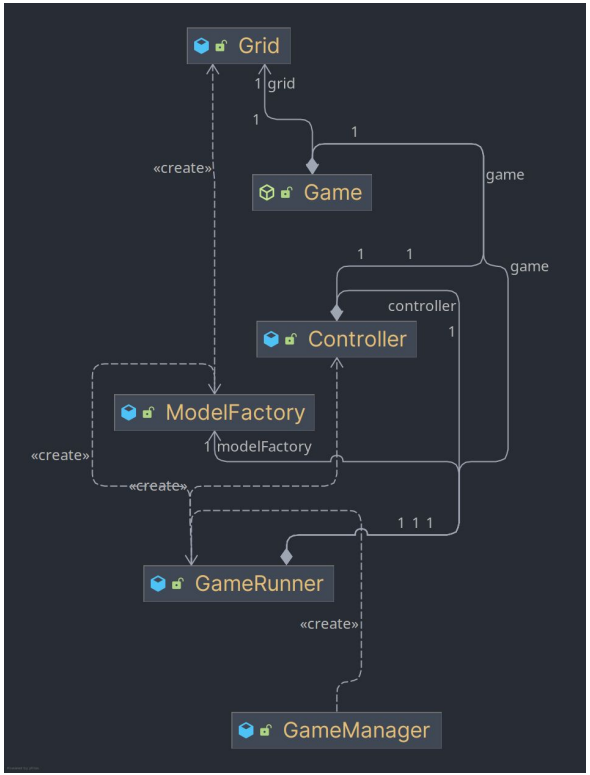
Changed Design - Language & Theme Handling







Feature Helped By Design: Multiple Simultaneous Simulations



Feature That Challenged Design: Buttons

